Grounding-First Industrial Safety Reasoning

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GitHub: https://github.com/billybirdlab2/capstone-agent

A Project description & goal

1 Problem statement and goal

I am developing an agent to improve the visual facts used to learn industrial-safety rules. Its first decision is which model, grounding configuration or semantic construction to modify when a scene description is not adequately supported by image evidence. The agent will compare candidate outputs, select a bounded action and iterate using independently computed feedback. Only predicates accepted by a separate verification step will enter downstream inductive logic programming (ILP) and safety-rule learning. The initial study is offline scene analysis, not autonomous robot control or safety certification.

2 Related work

Popper learns logical programs by using failed hypotheses to constrain subsequent search [1], while Meta Module Networks and Lu et al. demonstrate compositional visual reasoning and hazard-rule evaluation [2,3]. These methods motivate the rule-learning task, but reliable reasoning also requires correct perceptual inputs. Grounding DINO provides text-conditioned localization and Segment Anything provides prompted masks [4,5]; neither alone establishes a safety relation. ReAct motivates interleaving decisions, tool actions and new observations [6]. Building on my Lab 1 study of these reasoning methods, I will test adaptive repair of the grounding stage before attempting to improve rule induction itself.

B Why an agent

A missing object, an incorrect entity binding and a poorly defined relation can produce similar downstream errors but require different interventions. A single prompt cannot inspect the outcome of its own proposed intervention, and a fixed script follows the same ordering regardless of the failure evidence. The proposed agent chooses the next diagnostic action from a restricted set. I will test whether this flexibility is useful rather than assume that an LLM is better than scripted search.

Proposal: tools, evaluation and scope

Element	Commitment
Research question	Does adaptive selection of models and semantic/grounding configurations improve grounded-predicate correctness over an equal-budget fixed policy, and subsequently improve held-out safety-rule generalization?
Tools and independent check	Existing visual-model, detector and segmentation outputs; typed predicate construction; a planned verifier using reviewed visual labels, overlap metrics and binding checks. Literature retrieval supports decisions but cannot validate a scene.
Baseline and success	The same tool/action catalogue, inputs and resource budget with a fixed troubleshooting order. Compare predicate precision, recall and F1, false acceptance, abstention and cost; then hold the ILP learner fixed for the downstream comparison.
Milestones and main risk	November 15: one bounded, logged grounding-repair loop and fixed baseline. November 29: held-out comparison and ILP integration. The largest risk is unreliable relation ground truth; acceptance criteria must be fixed independently of the agent.

Current contribution: a working persistent research service and measured retrieval/citation infrastructure. The adaptive visual-grounding loop remains the proposed research experiment.

C Agent design

1 Workflow and information flow

One decision agent coordinates ordinary computational tools. Model inference produces candidate evidence; it is not an independent verification step. The grounding loop below is planned, while the literature tool and its citation checker are implemented.

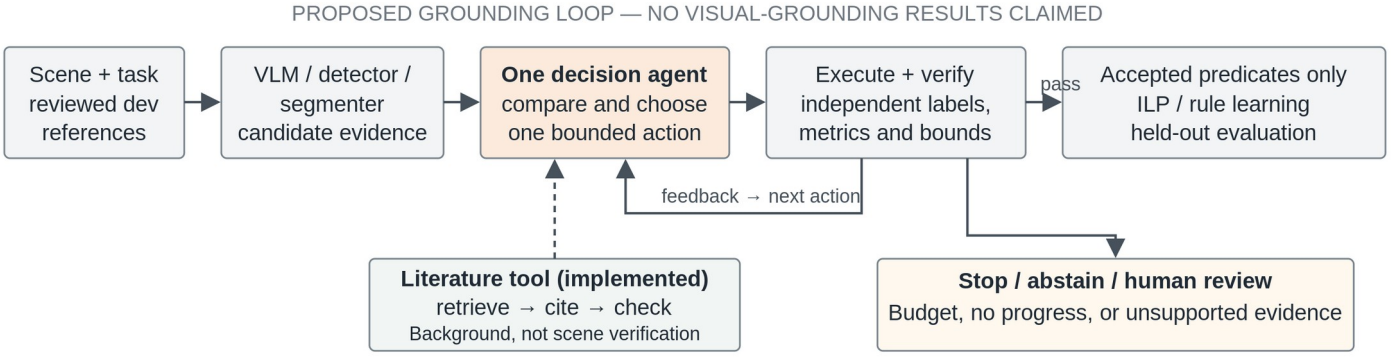


Figure 1. Literature informs action selection. Only the verifier releases accepted predicates; unresolved cases stop or escalate rather than entering rule learning.

2 Why each component is needed

Role / tool	Purpose and what is lost without it
Visual tools: VLMs, Grounding DINO, SAM	Propose descriptions, object regions and masks. Without them there is no visual evidence to compare; their outputs remain hypotheses.
Grounding decision agent	Selects a model, prompt, segmentation setting or semantic/predicate revision. Removing it yields the fixed-policy baseline, not a broken evaluator.
Executor + independent verifier	Enforces the action bounds and computes agreement with reviewed annotations. Without it, an agent could mistake confidence or an explanation for a successful repair.
Literature retriever + citation checker	Supplies method definitions and source-backed constraints. Without it, literature claims cannot be checked; it is deliberately outside scene acceptance.
Predicate gate + downstream learner	Passes only accepted facts with entity bindings and provenance into ILP. Without the gate, unsupported visual claims can contaminate learned rules.

Tool contract. A proposal contains scene/run IDs, the current configuration, one allowed action and its parameters, evidence references and a stop/escalation reason. The executor returns artifact IDs, metrics and resource use. Code rejects out-of-range actions; the agent cannot change labels, acceptance thresholds or verifier code.

Memory decision table (including Lab 5 Part E)

Stored information / kind	Location and key	Lifetime / control
Research papers / semantic	corpus/ + index/; document, section, edition and hash	Until the edition pin changes; read-only to the agent.
Approved facts and preferences / long-term	memory/MEMORY.md and USER.md; dated, sourced entry	Across sessions; agent proposes, owner merges.
Skills and policy / procedural	.claude/skills/, CLAUDE.md, PERMISSIONS.md, settings	Versioned; changes require owner approval.
Run notes / episodic	memory/daily/; date. Planned experiment log: run/action ID	Persistent audit trail; measurements written by tools.
Current hypotheses / working	Session state; scene, configuration and remaining budget	One run; held-out labels never enter agent context.

Retrieval is a consultation tool, not the research decision or a substitute for measurements. It is useful for model assumptions, failure taxonomies and predicate definitions; private experiment values must come from project logs. Retrieved text is treated as data, not instructions, and is not copied into persistent memory. Stop on verified acceptance, the fixed budget, a fixed no-progress limit, or insufficient evidence; escalate any needed change outside the allowed action set.

C Agent design — working demonstration

3 Persistent research service (Lab 4)

I configured a research digest covering VLMs/grounding, neuro-symbolic reasoning/ILP, robotic safety, robot learning/control and robotics systems. One cloud invocation searched five queries over seven days, requesting six results per query. The audited run returned 25 candidates and selected eight papers, wrote a dated digest and daily note, and pushed commit 80f62dc. I merged the digest through PR #3 and the dated project-memory update through PR #4 (memory commit 8e5ebb8). These demonstrate persisted repository state and owner-controlled acceptance. [E1]

Audit evidence	Observed result / limitation
Runtime and Git	Claude Code 2.1.316; model ID claude-opus-5-5. PRs #2, #5 and #6 record digest configuration, retriever installation and virtual-environment untracking; the evaluation branch is based on 69ebe45.
Permission tests	Probes of curl, pip install, a dry-run push to main and gh pr merge were denied. These test specific command patterns, not all possible ways to perform an operation.
Cloud workflow	The completed digest used Run now. The audited session lacked a PR-opening tool, so I opened its PR manually. A timer-triggered run and autonomous PR creation are not claimed.
Security limits	Server-side branch protection was unavailable for the private repository at the audit. The cloud allowlist/connector configuration still needs a current verification record; local policy is not proof of complete isolation.

Local research corpus and index (Lab 5 A–B)

The corpus contains 13 arXiv manuscripts, pinned to their first versions, spanning ILP, compositional vision, visual grounding, segmentation, VLM reliability and constrained policy learning. The manifest check reported agreement with all 13 files, and all 13 PDF hashes match the index lock. The repository is private; PDFs and extracted index text remain local pending confirmation of source permissions and the staff-access/rebuild route. The manifest's "licensed" labels are provisional classifications, not a completed license review. [E2]

Index / retrieval setting	Recorded configuration
Parsing and chunks	pypdf text parser; 350-word target, 50-word overlap; 801 chunks, no dedicated table chunks.
Embedding	Sentence Transformers; BAAI/bge-small-en-v1.5; model name and PDF hashes recorded in index/manifest.lock.
Keyword and hybrid	BM25: k1 = 1.5, b = 0.75. Reciprocal-rank fusion: constant 60, up to 20 candidates per retriever, final k = 5.
Execution	Python 3.12.10 in .venv. Exact installed dependency versions are not yet captured; minimum versions alone do not reproduce the full environment.

Agent tool-use and citation checks (Lab 5 C)

For a question about Popper's generate–test–constrain method, Claude executed retrieve.py and then check_citations.py with a structured answer, source metadata and quotations of at most 25 words each. The successful local session used .venv/bin/python3 for both scripts and reported PASS for chunks p020:03 and p024:03. A separate direct test accepted a genuine quotation and rejected a fabricated one. The checker validates quotation matching and metadata, not whether every answer claim follows from the quotation. [E3]

The corpus is parsed and embedded locally. Selected retrieved passages are passed to the Claude answerer, so local indexing does not mean that all answer-generation data stays on the Mac. The two script-specific virtual-environment permissions work locally but still need to be committed and documented consistently in the skill and permission specification. [E3,E4]

D Evaluation

1 Independent checks and measured retrieval results

I evaluated the fixed corpus and index using the course evaluate.py script. The 25 question texts were frozen before the retrieval pass: ten identifier, ten paraphrase and five intended not-in-corpus items. Claude proposed answers and chunk IDs from retrieved passages; an AI-assisted PDF review recorded source pages, sections and supporting text. Question q01 had an unsupported acronym premise and retains null gold fields. Therefore Hit@5 and MRR@5 use 19 answerable questions, including nine identifiers; complete student sign-off against the PDF pages remains to be recorded. [E2,E4]

Hit@5 is the fraction with at least one accepted gold chunk among five results. MRR@5 is the mean reciprocal rank of the first such chunk, with zero for a miss. These measure evidence retrieval, not answer correctness. Each null-gold item was also answered once through the model and deterministic citation checker in each mode.

Mode	Hit@5	Identifier	Paraphrase	MRR@5	Intended negatives
Embedding	11/19 (57.9%)	6/9 (66.7%)	5/10 (50%)	0.3632	5/5 abstentions
Hybrid	16/19 (84.2%)	7/9 (77.8%)	9/10 (90%)	0.5482	5/5 abstentions

The evaluator also groups q01 with the null-gold questions, so its raw abstention summary is 5/6 in each mode. The five intended negatives are q21–q25; q01 is discussed separately below.

Hybrid gained five net hits (+26.3 percentage points): it recovered q02, q11, q14, q15, q17 and q20, but lost q19. Both modes missed q06 and q08. Five of the six recoveries were paraphrases. Fusion therefore helped this question set, but not every query; no general superiority or downstream safety improvement is established. [E4]

The q01 case. The question asks for the expansion of “LFF,” which the corpus review did not establish in the Popper paper. Both model runs nevertheless supplied an inferred expansion with real quotations, receiving citation PASS rather than abstaining. q01 is retained as a question-construction error and a premise-handling challenge, not scored as a correct answer. Its behavior demonstrates that citation fidelity does not establish entailment. [E4]

Failure analysis. For q06, both modes favored general text/image-fusion discussion over the text-encoder passage. For q08, number-heavy flattened POPE tables competed with sampling-method prose. For q19, SpatialVLM’s joined words may weaken keyword matching; reciprocal-rank fusion can then demote a strong embedding-only passage. These are plausible mechanisms, not confirmed causal explanations: no parsing or keyword-only ablation was run. Some gold sets may omit alternative supporting passages. [E4]

2 Capstone baseline and success criteria (planned)

The research comparison will give the adaptive agent and a fixed troubleshooting policy the same model/action catalogue, scene inputs and resource budget. I will partition scenes by capture group before development, calibrate acceptance rules on reviewed development annotations, and keep held-out labels unavailable to the agent. A verifier will compute box/mask overlap, entity-binding validity and predicate precision/recall/F1, and release only accepted predicates. With the downstream learner held fixed, I will then compare held-out hazard precision, recall and F1.

Success means better grounded-predicate F1 at a predeclared minimum accepted precision and matched cost, without hiding abstentions or failures. Budget, thresholds and repeat count will be fixed before the held-out comparison. The present benchmark is small, has related questions and easy private-record negatives, and used one model sample per null-gold item. It does not establish repeated-run reliability or semantic accuracy on all 19 answerable items.

E Plan and risks

1 Progress and next milestones

Stage	Deliverable / evidence of completion
Completed	Persistent digest and owner merges; corpus validation and 801-chunk index; two-script citation demonstration; frozen questions, PDF source-review sheet and both retrieval-mode measurements.
By Nov 15 — Lab Report 3	One annotated scene batch through a bounded grounding decision, tool execution and independent check; typed action/result logs; fixed-policy baseline; recorded acceptance and budget specification.
By Nov 29 — Lab Report 4	Frozen-policy held-out and repeated-run comparison; accepted predicates handed to ILP; downstream rule evaluation; false acceptance, abstention, failure cases and resource costs reported together.

2 Main risk and question for the TAs

The central bottleneck is reference quality for spatial and safety-relevant relations. Agreement among models can reproduce a shared error, and a mask's overlap does not prove a relation. I would like TA feedback on the annotation and acceptance protocol, and on whether the equal-budget fixed policy is a sufficient baseline. Thresholds, resource caps and repeat counts are design decisions still to be finalized, not measured results.

Reproducibility and evidence

The working branch is lab5-evaluation, based on merged commit 69ebe45. The evaluations ran locally on October 31, 2026; exact environment capture, the two local permission additions and publication of the final evidence remain closeout tasks. A fresh-clone demonstration and staff access have not yet been confirmed. I will retain the input hashes, model pin and unchanged index with the results, and provide authorized source acquisition/rebuild instructions instead of assuming that a private repository grants redistribution rights.

[E1] Lab 4 Part H audit, October 31; digest commit 80f62dc, memory commit 8e5ebb8, PRs #3–#6. [E2] corpus/manifest.json; index/manifest.lock; questions and frozen-question record; PDF reference-review sheet. [E3] eval/transcript_C_retry.jsonl and direct checker-test output. [E4] eval/results_embedding.json, results_hybrid.json, corresponding console/runs files; reports/lab5-final-analysis.md and lab5-part-e-integration.md.

References

[1] A. Cropper and R. Morel. Learning programs by learning from failures. Machine Learning, 110, 801–856, 2021. doi:10.1007/s10994-020-05934-z; corpus: arXiv:2005.02259v1.

[2] W. Chen et al. Meta Module Network for Compositional Visual Reasoning. WACV, 655–664, 2021. arXiv:1910.03230.

[3] S. Lu et al. Neural-symbolic framework for dynamic hazard detection through compositional visual reasoning. J. Ambient Intell. Humaniz. Comput., 17, 979–994, 2026. doi:10.1007/s12652-026-05088-1.

[4] S. Liu et al. Grounding DINO: Marrying DINO with Grounded Pre-Training for Open-Set Object Detection. arXiv:2303.05499v1, 2023.

[5] A. Kirillov et al. Segment Anything. arXiv:2304.02643v1, 2023.

[6] S. Yao et al. ReAct: Synergizing Reasoning and Acting in Language Models. ICLR, 2023. arXiv:2210.03629.

Appendix — AI Use Declaration

I used Claude Code 2.1.316 (recorded model ID claude-opus-5-5) for tool execution, digest preparation, provisional reference answers, repository auditing and analysis. I used ChatGPT (GPT-6 Astra Pro) for setup/debugging assistance, question drafting, source-location review, numerical cross-checking, research-plan development and report drafting/polishing. The retrieval, indexing, citation-checking and evaluation scripts originate from the course; I configured and integrated them rather than claiming their algorithms as original work. I ran the reported commands, inspected test outputs and performed the owner merges. AI assistance was used to inspect PDFs; a complete per-question student confirmation record is still required. I retained q01 and its null reference rather than inventing a gold answer. All reported scores are measured outputs or arithmetic from those outputs; no grounding-repair performance is claimed. I remain responsible for the sources, interpretations and submitted work.